



Dissertation Handbook for Postgraduate Study

Course B31VZ ROBOTIC Masters Dissertation

The School of Engineering and Physical Sciences (EPS)

Electrical, Electronic & Computer Engineering (EECE)

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Dissertation Information

1. Introduction

To obtain the MSc degree, students must complete and pass a dissertation. You will be advised following the Semester 2 Academic Progression Board whether you have completed the PG Diploma stage of the program and can proceed to the dissertation stage. To be able to proceed to the dissertation a student must have met the minimum standards for progression, as referenced in the Programme Information (which can be found via your Student Portal).

The dissertation is the final stage of the master's degree and provides the opportunity to show that you have gained the necessary skills and knowledge to organise and conduct a research project. It should demonstrate that you are skilled in delineating an area, or areas, suitable for research; setting research objectives; locating, organising, and critically analysing the relevant secondary data, and authoritative literature; devising an appropriate research methodology; drawing conclusions; and if appropriate collecting and analysing primary data; making relevant and practical recommendations.

A dissertation is a 'formal' document and there are 'rules' that govern how it is presented. It must have chapters that provide an introduction, a literature review, a research methodology, results, and analysis, discussion, and, finally, conclusions and recommendations. Essays and descriptive situational analysis on their own are not acceptable as master's level dissertations; the structure and framework of the dissertation should be common regardless of type.

The master's level dissertation is distinguished from other forms of writing by its attempt to analyse situations in terms of the 'bigger picture as well as detailing 'what' can be done, it explains/explores the underlying 'why' The most successful dissertations are specific and narrowly focused, offering deeper and more reflective understanding.

This document is intended to guide you through the dissertation process. It can only offer suggestions; there is nothing that can be said that will guarantee the production of a fine piece of work, but these are suggestions only through time,

You must read this guide before starting your dissertation and consult it throughout the process. This will help you to complete your dissertation and make more effective use of your supervisory meetings.

2. Aims and Learning Outcomes

2.1 Aims

The aim of the MSc dissertation is for students to undertake independent research into a topic that is relevant to the practical pursuit of their subject. Students are expected to show evidence of independent inquiry and investigation.

Broadly, the aims of an MSc dissertation are to:

- design and conduct a substantial piece of independent research
- reflect critically on the relationship between theory and practice
- gather and sort data from a variety of written and electronic sources
- review, organise, and evaluate evidence and reflect and comment critically on it.
- engage critically with a wide range of complex literature.
- contribute in an informed way to debates and issues relating to the subject area
- identify and justify potential areas for research in the discipline
- use the information to support and substantiate claims.
- plan and write a substantial piece of work using academic apparatus and appropriately acknowledging the work of others.
- construct and present sophisticated, coherent, and persuasive written arguments.
- show evidence of ability to plan and manage a dissertation/project and work to a series of deadlines.
- exercise substantial autonomy and initiative in the research context.
- use a range of library-based (conventional and ICT) resources to source appropriate literature on the chosen topic.
- communicate research findings in an appropriate academic form.
- develop the ability to work independently where appropriate.
- use appropriate IT and presentation skills.
- (Where appropriate) enable interaction with practitioners; and
- (Where appropriate) manage ethical issues within the research context.

2.2 Learning Outcomes

After completion of the dissertation/project students should be able to:

- conceptualise a discipline-specific problem or research topic suitable for research within the specified degree subject level.
- critically evaluate a body of secondary data and authoritative literature to show that a clear research gap exists.
- define, design, and deliver an academically rigorous piece of research using appropriate research methods.
- implement an appropriate methodology to collect and analyse primary evidence.
- demonstrate the relationships between the theoretical concepts taught in class and their application in specific situations.
- evaluate how the evidence complements or contradicts existing studies.
- formulate evidence-based recommendations for practitioners and indicate where further research is needed.
- demonstrate the management skills needed to successfully a piece of research/project.
- show evidence of critical and holistic knowledge and have a deeper understanding of the chosen subject area.
- appreciate the practical implications and constraints of the specialist subject.
- understand the process and decisions to be made in managing a project within strict deadlines.
- set realistic research objectives and research questions that will close the research gap.
- contact, work with, and develop relationships with key research informants.
- work in a peer relationship (i.e., with dissertation supervisor) to manage and improve individual performance.
- take significant responsibility for organising a range of resources; and,
- (Where appropriate) manage complex ethical issues within their research context.

3. Styles

Dissertations need to demonstrate knowledge and understanding beyond the undergraduate level and should also reach a level of scope and depth beyond that taught in class. All dissertations must be presented in an appropriate academic style and format to ensure that the precise aims of the dissertation are met. It is important the aims and objectives of the dissertation must be eased and are achievable within the scope of the dissertation framework. Academic style does not only refer to the clarity of expression, grammar, use of citation, and, referencing but relates to a structured approach to justification action and validation of facts, theories, and opinions presented to form a precise argument. With the range of subjects taught across the school, the type of research dissertation will vary. Your Dissertation Coordinator/Programme Director will give you specific guidance.

4. Key Contacts and Dates

4.1 Contacts

Dissertation Supervisor: Your dissertation will be written under the guidance of a Dissertation Supervisor who will normally be a member of the academic staff knowing et the arched and/or general research experience in the context of your degree discipline. Your supervisor is there to guide you through the dissertation process. They will not necessarily be experts on the specific topic you are researching but they will have an exceptional understanding of the research process. They will challenge your ideas to help you refine them, warn you about common pitfalls, and listen to your plans.

Dissertation Co-ordinator: The Dissertation Co-ordinator has overall responsibility for coordinating including administration of courses, responsibility for CANVAS, allocation of students to supervisors, arranging workshop sessions as appropriate, and ensuring that the regulations and procedures outlined in this booklet and the School Engineering and Physical Sciences (EPS) Postgraduate Student Handbook are followed. You should normally only contact the Dissertation Co-Ordinator if you are unable to contact your supervisor.

EPS Research Ethics Committee: Research associated with dissertation courses must satisfy the ethical practices observed by the School's Research Ethics Committee.

4.2 Dates / Timescale

The following is based on a Full-time Master's student who begins their program study in Semester 1 of the academic year (note: exact dates will be notified to you via CANVAS):

Jan-April	Pre-requisite Research Methods course which includes developing ideas, developing objectives, and completing an initial review of literature, workshops, and information sessions. • You will be allocated a supervisor in Semester 2 • You will develop your research proposals during this period for submission typically in week 12 of the Research Methods mandatory course B81EZ
March/April/May	Submission of Semester 2 coursework, Exams, etc
May/June	The academic Progression Board approves the progression to the dissertation. If you are required to resist any exam in the August exam diet to progress, you will not start your dissertation research until September.

June/July	Submission of Ethical Approval Form (in consultation with the supervisor): The form must be submitted before primary data collection
	May/June Progression: post Semester 2 diet deadlines
Early August	Poster / Presentation Early -Mid-August – Information will be published on CANVAS
Last week in August	Dissertation submission – deadline will be published on CANVAS

5. Supervision

You will be supported through the dissertation by a dissertation supervisor who will be qualified and approved to supervise dissertations at this level, regardless of the topic. Where possible, dissertation supervisors may be allocated based on subject expertise in an area like that being studied. This will not always be possible, and you may be allocated a supervisor with more general subject knowledge. Where more specific subject guidance is deemed necessary by yourself and your supervisor, your supervisor will liaise with appropriate, available subject specialists to help inform your supervision. The extent of that subject expert's input will be limited to advice concerning the basis of your literature review or data analysis.

Before the allocation of a supervisor, it is recommended that students visit members of staff on an informal basis to explore their ideas. Staff members are not obliged to meet with students for this purpose but most staff members in the school are amenable to short meetings to discuss direction. It is recommended that you make use of established office hours where possible. Where you are studying courses in your preferred chosen/pathway, it may be useful to talk to the coordinators of these courses for guidance and, of course, discuss your ideas with your mentor and/or program director is that you will go into any discussion with ideas to be challenged and not with an expectation that someone else will provide you with ideas.

5.1 Role of supervisor

The dissertation supervisor performs many functions but the responsibility for the content and, hence, quality of a dissertation is entirely that of the student.

It should be emphasised that the submitted dissertation is entirely your work. You ask your supervisor, however, to read in detail a draft of a portion of your dissertation normally up to a maximum of two

chapters (or equivalent), to give feedback on presentation, content, and style. Advisors may comment on chapter outlines and may scan through options discretion

The dissertation supervisor will **NOT** read the finished work before submission and will not comment at any stage on the expected grade or mark.

The dissertation supervisor will not be expected to check or correct grammar, spelling, or expression.

If during the preparation of the dissertation, the focus and direction of the dissertation change substantially from that outlined at the outset, then you should immediately seek approval for this change from your program director and dissertation supervisor and provide evidence of your work.

The supervisor role includes the following:

- To advise the student whether the project appears to be feasible and the possible risks that may be involved, for example, problems in trying to access information, and potential poor response rates to surveys concerning commercially sensitive issues.
- To assist the student in understanding resource constraints.
- To challenge the student in determining the validity and relevance of ideas, reading materials, and appropriate academic frameworks within which the candidate seeks to place the topic.
- To explore the choice of methodological approach(es).
- To monitor progress and to advise on what is required to achieve a satisfactory dissertation.
- (For approved company-based dissertations only) to liaise with the company supervisors and resolve any problems the student may have in obtaining access to company information or personnel.
- To assess the dissertation (as '1st Marker') and to submit a report.

5.2 Responsibilities of the student

A supervisor will expect students to do the following:

- To supply a research proposal for the intended dissertation.
- To select a topic that is appropriate for their degree program
- To plan their research project and track their progress. If a student is experiencing difficulties in progressing this must be communicated to the supervisor at the time they are encountered. Retrospective information is not acceptable.
- To listen to and respect the guidance offered by the supervisor.
- To maintain regular contact with the supervisor either by e-mail, MS Teams or F2F
- Students are expected to take a proactive maintaining contact and bring material or options to be discussed rather than expect their supervisor what should be done next.

- To attend supervisor meetings when they are booked. Supervisors expect students to inform them if they are unable to attend any scheduled meeting. Supervisors are under no obligation to rearrange missed appointments unless good reasons are given by the student.
- To write in clear English using appropriate academic terms and citation and referencing conventions. It is not the responsibility of the supervisor to ensure that this condition is met.
- Students who are conducting a company research project are expected to maintain regular contact with the company and appropriate representatives. Students must be mindful that they are representing Heriot-Watt University during this period and conduct themselves according to the highest standards of professionalism.
- To inform the supervisor of any absence (sickness, personal, family visits, holidays, work experience) during the time nominated for working for the completion of the Dissertation
- To follow the focus and direction of the research project as outlined in the Research Proposal Form.
- To take and maintain reliable notes from discussions with supervisors. Psychology students, you will be required to document these within your lab book.
- To write the dissertation with guidance from the supervisor. Work must be the student's own. The dissertation reflects the student's work and research abilities, not the supervisors.

5.3 Supervisor Feedback Meetings

Students are allocated a maximum of five formal feedback meetings with their academic supervisor. The purpose of these feedback sessions is to discuss progress and resolve any difficulties. Students are expected to take a proactive approach and bring material or options to be discussed rather than expect the supervisor to say what should be done next. Students are responsible for arranging feedback sessions with supervisors.

Feedback sessions may include any of the following forms of communication to face meetings with your supervisor.

1. Remote meetings with your supervisor (e.g., via Skype, MS Teams)
2. Telephone conversation using Supervisor's office telephone landline number.

Additional email correspondence may be required to set up meetings and to confirm or clarify any advice given during a formal meeting. If for any reason all the listed forms of communication above are not possible for a scheduled meeting, then the email may be used as a last resort.

The supervisor will endeavour to discuss your research with you as soon as possible but you must remember your supervisor has other work commitments, conferences to attend and the right to have a summer holiday— please ensure you build this into your schedule.

A record is kept of each of these feedback sessions detailing the dates, discussion content, and any action points. This may be written by the supervisor or the student but needs to be 'agreed'.

6. Ethics

Every student proceeding with a dissertation is required to fill in an ethics form. It is the student's responsibility to ensure that this form is completed well in advance of the fieldwork.

No dissertation will be allowed to proceed which specifically seeks to conduct primary research amongst participants under the age of 16 or over the age of 65 without the specific support and agreement of the supervisor and ethics committee. Furthermore, any dissertation that seeks to undertake primary research amongst any potentially 'vulnerable' participants is highly unlikely to gain ethical approval without adherence to the university's policies on research engagement.

7. Proposal

The final dissertation proposal is prepared after discussion with your supervisor. It is an important working document that will evolve into the dissertation over the writing period.

Used correctly, the proposal will become your road map through the dissertation process. Because of a wide variety of circumstances, the focus of your research may change. If this happens, you must agree with your dissertation supervisor that such a change is appropriate.

The Dissertation Proposal shows that you have been through key issues, such as the main research aims and objectives, the main sources of primary and secondary data, the key literature of relevance, and the research methodology. The Dissertation Proposal should provide your dissertation supervisor with a 'detailed skeleton' of the whole dissertation.

8. Approval of Data Collection Instruments

All data collection instruments i.e., questionnaires, interview questions, experiment designs, and focus group topic guides that a guide in research must first be approved by your dissertation supervisor and, in the case of the company research project, by the company concerned. This is to ensure that questions meet your research aims and are appropriately framed. The dissertation supervisor will not make the necessary corrections but highlight where improvements are required. See also Section on Ethics.

9. Plagiarism

Plagiarism involves the act of taking the ideas, writings, or inventions of another person and using these as if they were one's own, whether intentionally or not. Plagiarism occurs when acknowledgment that the writings or ideas belong to or have come from another source.

The University takes plagiarism very seriously, each year some students are unable to graduate because their work is being investigated by the University Academic Standards Disciplinary Committee worked through Turnitin which will let you see the similarity index reports. In some cases, Turnitin may not identify a similarity that does exist; however, this is likely to be picked up by experienced academics. The best way to avoid plagiarism is to explain ideas in your own words and phrases; it may not sound as fluent, but it will demonstrate your understanding of the subject far better than using someone else's words.

Most plagiarism is unintentional or accidental. It is therefore vital that you are aware of common pitfalls which can lead to suspicions of plagiarism.

Plagiarism comes in many forms, but typical examples include:

- *Collusion* – when one student allows another student to copy their work.
- *Complete plagiarism* – where the work is copied entirely from one or more sources.
- *Partial plagiarism* – this involves inserting an unacknowledged passage of text into the coursework.
- *Word switch* – copying sentences and changing a few words without appropriate reference to the source.
- *Concealing sources* – failure to make ongoing acknowledgments to sources used sources and once in a piece of work.
- *Self-plagiarism* – when a student uses material from a previous assignment in a current assignment.
- *Inadvertent plagiarism* – the result of poor study skills.

Use of EndNote Online (see below) can help you ensure that you cite sources correctly.

Dissertation students must be particularly careful about self-plagiarism. If a piece of work has been submitted for a course such as Research Methods, it cannot be submitted without change for the Dissertation. That piece of work has already been used to gain credits. Students can use the Research Methods work as the basis for their Dissertation but are expected to add to the work, provide evidence of further reading, be more critical of literature, and provide fuller discussion.

The university's guide to plagiarism can be accessed using the following link:

[Heriot-Watt University \(hw.ac.uk\)](http://www.heriot-watt.ac.uk)

10. Format of Final Submission

10.1 Length

The length of your dissertation will vary depending on your program of study:

- Dissertations should be between 12000 and 15000 words
- Word counts are including text notes but exclude cover page, personal statements, acknowledgment, abstract, table of contents, list of abbreviations, reference list, and appendices.
- All dissertations must include a word count which must be shown on the title page of the dissertation. A dissertation is not validly submitted until it is accepted by the Dissertation Coordinator as being within the word count limit.

10.2 Writing Style

The dissertation is a 'formal' document and, as such, its contents must be expressed in a certain style. That style is the third-person singular passive. For example, consider the following statement:

10.3 Formatting

The dissertation must follow the following requirements:

Aspect	'Word' Command Line	Recommended Setting
<i>Language</i>		English (UK)
<i>Word count</i>		As above in Section 10.1 Excludes abstract, references, tables, and graphs i.e., main text only
<i>Paper Size</i>	File, Page setup – Paper.	A4
<i>Margin Settings</i>	File, Page setup, margins	Top 2.54cm Bottom 2.54cm Left 3.17cm Right 3.17cm Gutter 0cm Gutter Position Left
<i>Line spacing</i>	Format, Paragraph, Indents & spacing	One-and-a-half spacing, not single or double.

<i>Page Numbering</i>	View, Header & Footer, Insert page Numbers	centre
<i>Text</i>	Format, Styles & Formatting	Times New Roman
<i>Font</i>	Format, Styles & Formatting	12-point font
<i>Alignment</i>	Format, Styles & Formatting	Justified (i.e., straight edges)
<i>Typical Dissertation Layout</i>		
Front cover		Required
Acknowledgments		Required
Abstract		Required
Table of Contents	Insert, Reference, Index & Tables	Required
List of Figures	Insert, Reference, Index & Tables	Required (Note all figures in the main text must be numbered, titled, and attributed)
List of Tables	Insert, Reference, Index & Tables	Required (Note all tables in the main text must be numbered, titled, and attributed)
Abbreviations		Optional
Glossary of Terms		Optional
Main text Chapter & Section Headings	Format, Bullets & Numbering, Outline numbered	Required, The chapter title, bold, outlined numbered e.g., Chapter1, Start each chapter on a new page. Section title bold outlined numbered e.g., 1.1, Subsection title, outlined numbered e.g., 1.1.1
References		Required Harvard system is preferred but any other consistent engineering style is permitted (discuss with your supervisor)

Appendices		Optional Appendix title, bold, outlined numbered e.g., Appendix 1, start each Appendix on a new page
Format		Word or pdf

11. Dissertation structure

The dissertation must be divided into appropriate chapters and sections. The following elements are required, if you intend to depart from this format, you should discuss the overall structure with your supervisor.

- Title Page (**refer to the template on page 30**)
- Abstract
- Acknowledgement
- Authority of ownership (**refer to the template on page 31**)
- Contents Page
- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Methodology
- Chapter 4 Findings/Data Analysis
- Chapter 5 Discussion
- Chapter 6 Conclusion
- Chapter 7 Recommendation
- References
- Appendices

The following section provides more detail on what is expected within each section.

Chapter Content

Each of your chapters will require both an introduction and a conclusion. The former provides the reader with a content 'map' of what is to come, and the latter provides a concise summary of what they have just read. Each introduction should look back to the conclusion of the previous chapter, and forward to the contents of the chapter which you are introducing. These conclusions and introductions act like small links that bind the 'chain' of the chapters together in a more seamless whole than would have occurred if the chapters had not been introduced or concluded; they 'smooth out the transition from chapter to chapter and from topic to topic.

Title Page

The title page must contain the student's name and enrolment number, year of graduation, supervisor's name, word count, and the following wording: "Dissertation submitted in partial fulfillment of the requirements for the degree of MSc at Heriot-Watt University". **(see the template on page 30)**

Abstract

No more than 350 words are to be completed after the dissertation/company report has been completed. An abstract summarises the following:

- Research Aims The main themes and ideas for areas of the theory being investigated.
- Boundaries: The background and context of the research.
- Research Questions: The questions that informed data collection.
- Methodology: The main method(s) used to collect data and the sample.
- Results: A summary of the answers to the research questions.
- Conclusions: The conclusions formed from these results.
- Recommendations: Recommendations for future research and practice.
- Main research keywords must be included at the bottom of the Abstract

Acknowledgment

Mandatory for all students, a brief statement, signed by the student, should be provided which:

- Gives an assurance that the submission is the student's work (see Section 10 regarding plagiarism).
- Acknowledges that the primary data collection gained ethical approval for the study via the Chair of the School's Ethics Committee.
- Gives thanks to those who have given direct assistance through the course of the project. Try to keep more personal comments about family and friends to a minimum.

[Contents Page](#)

The contents page should list the chapter headings, appendices, references, and the respective page numbers. Separate listings should be given for lists of figures, tables, and abbreviations. You will find this easier to create if you know how to generate these lists using the indexing and style functions within Microsoft Word. Please note that there are courses available via the Herriot Watt Library to help you in this area.

Chapter 1 Introduction (Typically, approx. 500-1000 Words) The introduction should clearly state the aim of the dissertation and provide a clear rationale for the choice of the research topic. It should be clear why this topic is important to your program subject area theory and/or practice. There should be a straightforward evidence-based account of important trends, definitions, or pivotal ideas that underpin the choice of topic.

This short concluding section of the Introduction should indicate the content of each of the following chapters and how the chapters are related to each other. You are providing the reader with a 'road map' of the work ahead. Thus, immediately, they can see (1) where they are starting from, (2) the context in which the journey is taking place, (3) where they are going to end up, and (4) the route they will take to reach their destination. Such a 'map' will enable the reader to navigate their way through your work much more easily and appreciate to the maximum what you have done

Chapter 2 Literature Review (Typically, approx. 4000-6000 words).

The literature review shows your skill in locating and critically evaluating the sufficiency of secondary data in the form of published academic articles and practitioner surveys. Your critical evaluation should reveal the research gaps that motivated you to plan primary research on the topic.

After the literature review, you should clearly state your research aim, objectives, questions, and/or model for testing with an indication of hypotheses.

The content of a literature review may be drawn from all, or a combination of, textbooks, journal articles, conference papers, reports, case studies, the Internet, magazine features, or newspaper articles. It should be remembered, however, that the most important source of academic literature is journal articles, and you should ensure that you are familiar with the most recent publications in journals relevant to your subject area.

The critical evaluation should provide the foundation for your research aim, objectives, and questions/hypotheses. Remember 'critical' does not mean looking at the negatives but, also highlighting the positives to form an evaluation.

To complete a literature review you must select appropriate search terms, use these in appropriate databases, read a range of materials, summarise the contents, evaluate the sufficiency of the knowledge, and synthesize (combine) your evaluation to highlight the research gap.

You must demonstrate that you have found a range of appropriate literature. The articles that you locate should be both seminal articles that detail foundational concepts and current research that updates these concepts.

A literature review enables you to demonstrate that.

- You discovered and reported what you have found to be relevant in the literature search that you have understood it and that you can analyse critical manner.
- Your knowledge of the area of interest is detailed enough that you can identify gaps in the coverage of the topic, thus justifying the reason(s) for your research.
- You have identified a dependent variable (outcome/result), key independent variables (influences), key trends and 'actors' in the environment of your study.

The literature review should provide a clear justification for the research objectives and questions. It should not be a descriptive catalog of authors, frameworks, and ideas but should attempt to introduce a critical evaluation of those authors' work.

A good literature review will mean that, later in your dissertation, you can draw on the literature in the methodology and analysis chapters to argue for the validity of your choice(s) of research approach, the appropriateness of the process by which you analyse your results, and again in the discussion when you identify whether or not your findings are congruent with the accepted research which has gone before.

Students are permitted to develop the same topic that they identified in the pre-requisite course, however, are strongly advised to identify and analyse additional literature. The reasons for this advice are specifically because:

- To simply copy previous 'assessed' work would constitute "self-plagiarism" and is an unacceptable academic practice. You have been credited for this work already!
- The Dissertation is the culmination of your studies. As such, it is expected that it will be developed from the work you completed within the pre-requisite course (where you were "getting to know" your ideas and practices).
- There is ample time to develop a substantial literature review within the timeframes of this course.

Chapter 3 Research Methodology (Typically, approx. 1500 to 2000 words).

- The Research Methodology commences by restating the research objectives and details of the precise research questions of the project. Your methodology is assessed on whether it is a valid approach to answering these questions. Students should identify the choice of philosophy and method, the reasons why that choice was made, and the pros and cons of the justification should be based on the research question and context and not primarily on concerns relating to time and cost. A good methodology will painstakingly argue for and justify each decision that is taken when arriving at the research design. Depending on the discipline area, the chapter should contain information.
- the limitations of secondary research (the gaps identified in the literature review),
- the research philosophy
- the research strategy (qualitative or quantitative methods)
- detailed discussion of data collection instrument (interview topic guide, survey, etc.)
- population and sampling approach
- research instrument administration (how you collected the data)

- data inputting
- data analysis plan
- ethical issues that were considered and how addressed
- what constitutes the sample
- the sampling criteria used and
- the response rates

The conclusion of this chapter should provide a summary of the main points that have been covered. The conclusion should also direct the reader as to how the contents of this chapter link in with the contents of the next chapter, your findings.

Chapter 4 Findings/Data Analysis (Typically, approx. 2500 to 4000 words)

The chapter should present the primary data you have gathered. This may be qualitative or quantitative data. You should describe the nature of the sample and the defining characteristics that make the participants of interest to your study. Any steps that you take to establish the validity and reliability of your data should be given. Any problems in the execution of the research project and their occurrence should be discussed and explained (this does not devalue your work but shows criticality).

After describing the sample, you should proceed to analyse your data so that you clearly show how it relates to your research questions/ hypotheses.

Please note that you may need to supply evidence of how you arrived at your findings. You should retain copies of all fieldwork and analysis should they need to be consulted by internal examiners.

You should clearly show the ability to reduce a large amount of data to interpretable information. This data must be organised logically and coherently ordered whole so that your thought processes and interpretation are clear to the reader.

You need to present your results in an accessible manner to show what the research has uncovered and to include only the most pertinent statistics or interview quotes as core evidence.

Whatever form of data analysis has been undertaken, it must be accomplished with care and attention to detail, as should how the results are presented.

For qualitative data, identified quotations should be included as evidence to support the conclusions from the data analysis. It is advisable to focus on the points of agreement and disagreement that emerged during interviews. You should avoid lengthy quotations unless they are of critical importance. However, short excerpts enrich the reader's understanding of the issues and provide you with the opportunity to shed a clearer insight on the topic. Tables and figures can be used to break up text and show the relationship between constructs.

For quantitative data, summary statistics should be used i.e., means, and frequencies. Statistics should be presented to no more than two decimal points to aid understanding. You should convert SPSS output to Excel tables and give a full indication of questionnaire items and not abbreviated labels.

The choice of methodological approach may affect the presentation of data. For instance, quantitative data presentations may require the use of fewer words. This may allow more words to explain the quantitative method within the methodology section. Qualitative presentations may require a particular framing of data across multiple sections (e.g., combining findings and discussion chapters). These are issues that you should discuss and clarify with your supervisor.

Chapter 5 Discussion (Typically, approx. 1000 to 1500 Words)

In the chapter, the introduction states the research aim, objectives, and questions. In this chapter, you relate the findings to each question and should highlight the most pertinent and important results.

The next step is to critically discuss the findings concerning the wider research literature. What this means is looking at where your findings are consistent with theory and where they contradict the theory. This means that you will draw links between your research findings and other published work – you may need to refer to your literature review.

This is the heart of the dissertation and must be more than descriptive. This chapter develops analytical and critical thinking on results and analysis concerning theoretical arguments grounded in the literature review.

Chapter 6 Conclusions (Typically, approx. 1000 to 2000 Words)

This is the conclusion of the whole dissertation. In this chapter you summarise what has been learned from the entire process, noting the specific contributions of your work to knowledge and practice (taking care to identify key audiences for your work). This will include the research gaps, the methods used for data collection, and the findings. Your summary will bring together the work of the

dissertation by showing how the initial research plan has been addressed in such a way that conclusions may be formed from the evidence of the dissertation company projects. No new material or references should be placed here.

After you have summarised the nature and value of your dissertation content, you should conclude by discussing the implications for two key audiences (1) Practitioner (e.g., how will your insights affect their understanding; how might practitioners utilize this understanding based upon your evidence? (2) Academic (how will your insights affect understanding of your subject area (e.g., topic; theories; concepts; etc.) and how might academic audiences utilise this insight based on your evidence?

You should discuss the limitations of your research and make suggestions of how further research might address these limitations – be mindful that you remain consistent with your chosen research philosophy. Remember that limitations aren't always problems! Rather, they set the parameters for understanding the value of your work. Be careful not to make claims that are not substantiated by the evidence you have presented in earlier chapters. You should also consider how research in this subject area could be pursued in the future – this will demonstrate your understanding of how your study contributes to the development of the overall body of knowledge.

References:

The University has a policy that covers all master's students about a reference system. It is important and you must get citations and references correctly. You must always cite the source of your material; inadequate citation could leave you open to accusations of plagiarism.

All references used in writing the dissertation/company project (whether direct quotations or paraphrasing) should be included in a reference list, compiled in alphabetical order by the author. The Harvard system for listing references or a consistency engineering referencing style should be used.

<http://www.hw.ac.uk/is/docs/Harvardguide.pdf>

Where there is any formal mention that you make in your company research project or master's dissertation for something written by someone else you must use in-text referencing. Every citation must be supported by a reference at the end of the report which supplies the details which will enable the reader to follow up on that citation.

The reference section is placed at the end of the company research project or master's dissertation (in alphabetical order of the first author's surname), as follows:

Text notes can provide additional information on points made in the text of the company research project or master's dissertation. They may be presented as footnotes or as endnotes, either of which should be kept to a minimum. Neither footnotes nor endnotes should be used, however, if the sole purpose is to give a page or reference, but these should be presented as parenthetical insertions into the main text.

Appendices:

Appendices may be used to provide relevant supporting evidence for reference. Students may wish to include in appendices, evidence that confirms the originality of their work or illustrates points of the principle set out in the main text e.g., questionnaires, and interview guides. Only subsidiary material should be included in appendices. Students should not assume that Appendices will be read by Examiners in detail.

If used, a copy of one interview/focus group transcript should be included in the appendices, and a copy of the questionnaire or focus group guide. External Examiners can also ask to see evidence of the research undertaken – you are asked to keep a copy of all recordings you undertook as part of your data collection (and you should include these as raw data on your memory device at submission). Many research organisations require data to be accessible for a period of up to 3 years after collection.

Appendices should be restricted to no more than 25 pages. You can keep additional information in a folder for use by the markers if requested.

12. Confidentiality

If the material in a dissertation or company project is regarded as confidential, the student should mark the text 'CONFIDENTIAL' on the front cover when it is submitted. In this event, it will be seen only by members of the Board of Examiners and administrative staff, and it will be kept secure, away from those available for consultation. No one outside the Board of Examiners and administrative staff will be allowed to see it unless they are given written permission by the student and Programme Director. Requests to see such company research projects or dissertations are almost universally refused. Such company research projects or dissertations will be kept secure for 2 years only unless the student specifically requests a security classification to be applied for a longer period.

13. Submission

All dissertations are to be submitted electronically via Turnitin on the relevant CANVAS course, by the published submission deadline. No hardcopy submissions are required.

‘Surname First Name_Supervisor’s Surname’

All students are required to submit any supporting electronic data, e.g., Excel files, Charts, sound files from interviews, transcribed interviews in Word process format, re-analysed data, etc. via One Drive [see Appendix 2 for instructions]. It is your responsibility to ensure that submitted electronic files are readable. You must keep a copy of all data, including that which cannot be submitted in an electronic format. Examiners or other School officials may request such data. Field data includes, for example, completed handwritten questionnaires and interviews, diaries, non-digital audio recordings, etc.

Please ensure that you submit it before the deadline (do not leave it until the last minute) in case you experience any technical difficulties. Technical difficulties are recognised as valid Mitigating Circumstances.

In line with the University’s Submission of Coursework Policy:

- No individual extensions are permitted under any circumstances unless the Dissertation Course Leader or Programme Director gives an extension to an entire class.
- A standard 30% of the mark awarded will be deducted should the work be submitted late, up to a maximum of five working days after the deadline date and time.
- In the case where a student submits work up to five working days late and they have valid mitigating circumstances, the mitigating circumstances policy will apply, and appropriate mitigation will be applied.
- Any work submitted after five working days of the set submission date shall be automatically awarded a no grade with no formative feedback provided.

It is vitally important that students report any problems that have affected or will affect their performance on the dissertation as soon as possible to their supervisor. You should as a minimum, contact the school office and submit a mitigating circumstances form. The Examiners can and will take mitigating circumstances into account, but only if they know about them before the Examination Board meeting. Please consult the student handbook for details of submission of evidence of Mitigating Circumstances.

13.1 Deadline

Exact dates will be confirmed via the relevant CANVAS course, however typically:

- Full-time students: late AUGUST
- Part-time students, and any full-time students who were required to revise and resubmit dissertations following submission in August: late NOVEMBER.
- Students who were required to resist a taught course before progressing to dissertation: early DECEMBER.

14. Marking

All dissertations will be read and assessed by two internal examiners. For company-based dissertations, the company will not be involved in the assessment process. In addition, some dissertations may be read by an External Examiner to ensure a universal standard is maintained.

- An agreed mark will be awarded for the dissertation. The following common standards apply:
 - The dissertation must be presented using a coherent and thoughtful level of English.
 - Clear analysis of the information collected should be included and this must be directed towards answering the research questions raised by the dissertation. An informed description of events or data is not enough.
 - The dissertation must show an awareness of the relevant literature.
 - The document should demonstrate the author's (i.e., your own) contribution to specific and wider understanding.
-
- **In addition, the assessment criteria used in the assessment will reflect the following:**
 - A depth of knowledge and critical understanding of an interdisciplinary or specialist area that goes beyond the final year undergraduate level and builds upon the student's postgraduate studies.
 - An ability to bring together coherently the perspectives of two or more theoretical standpoints and apply the results in a practical setting.
 - An ability to critically appreciate, to a higher level than that of a final year undergraduate student, major issues and problems internal to the discipline and/or regarding the discipline's impact on the external world.

The Examination Board has the right to require a viva-voce before awarding a mark. Usually, this is only required in exceptional circumstances.

A mark of 50% (Grade C) or greater is required to pass the dissertation. When a mark of 40-49% (Grade D) is awarded, this is classed as an initial failure, but with the possibility of revision (i.e., the dissertation should be revised and resubmitted and the resubmission mark is capped at 55%). This applies where only minor revisions are necessary to bring the work up to the required standard and resubmission is at the discretion of the Board of Examiners. Candidates are provided with specific guidance concerning the required changes (see later section of this handbook).

- *Minor editorial corrections or deficiencies:* Minor corrections of English, spelling, statistical errors, poorly justified or described methodology, but with adequate data; some reorganising of materials in the literature review, or results or findings, but not all sections fall under this heading

Where a dissertation has major deficiencies, it will receive a mark of less than 40%; in such cases, resubmission is not permitted, and students will be awarded a Postgraduate Diploma.

- *Major deficiencies:* This may reflect a literature review limited by the extent of materials and ideas reviewed or limited by the understanding being shown; inappropriate or inadequate research design; inaccurate or inadequate data analysis; lack of conclusions and understanding. Dissertations in this category would require substantive improvements, introducing a substantial number of new materials within several chapters.

To achieve 80% + (Distinction) a dissertation will:

- The report is publishable
- seek to attain a clear and explicit set of aims and objectives which are entirely appropriate to the study and whose fulfilments are entirely feasible within the scope of the study
- display a structure that is entirely clear, logical, and orderly, given the aims of the study, and which involves no omissions or digressions
- select an entirely appropriate body of data, in qualitative and quantitative terms, which is consonant with the aims of the study
- show a full and detailed understanding of a range of relevant theories and theoretical concepts
- demonstrate a clear ability to critically evaluate secondary sources.
- demonstrate a clear ability to establish a theoretical framework that is entirely the purpose of the study.
- present an in-depth and insightful analysis of the chosen data which is underpinned by the chosen theoretical framework, and which allows for the achievement of the stated aims of the study
- demonstrate unmistakable evidence of independent thinking
- be clearly and coherently expressed in an appropriate academic register
- Adhere fully to academic conventions in terms of referencing and presentation

To achieve 70% + (Distinction) a dissertation will.

- seek to attain a clear and explicit set of aims and objectives which are appropriate to the study and whose fulfilments are feasible within the scope of the study
- display a structure that is clear, logical, and orderly, given the aims of the study, and which involves no major omissions or digressions
- select an appropriate body of data, in qualitative and quantitative terms, which is consonant with the aims of the study
- show a thorough understanding of relevant theories and theoretical concepts
- demonstrate an ability to critically evaluate secondary sources
- demonstrate an ability to establish a theoretical framework that is appropriate for the study
- present a relatively detailed analysis of the chosen data which is underpinned by the chosen theoretical framework, and which allows for the achievement of the stated aims of the study
- demonstrate evidence of independent thinking
- be clearly and coherently expressed in an appropriate academic convention in terms of referencing and presentation

To achieve 60% + (Pass) a dissertation will:

- seek to attain a set of aims and objectives which are generally appropriate to the study
- display a structure that is generally clear, logical, and orderly, given the aims of the study, and which involves few major omissions or digressions
- select an appropriate body of data that is generally consonant with the aims of the study
- show a thorough understanding of relevant theories and theoretical concepts
- demonstrate an ability to evaluate secondary sources
- demonstrate an ability to establish a theoretical framework that is appropriate for the study
- present an analysis of the chosen data, which is underpinned by the theoretical framework, and is geared toward the achievement of the study's aims
- demonstrate some evidence of independent thinking
- be well expressed in an appropriate academic register
- Adhere to academic conventions in terms of referencing and presentation.

To achieve 50% + (Pass) a dissertation will:

- seek to attain a set of aims and objectives that are appropriate to the study, but which may not be comprehensive and/or may include some aims which are out of the scope of the study.
- display a structure that is generally logical but may involve omissions and/or digressions!
- select a body of data that allows at least partial fulfillment of the study's aims
- show some understanding of relevant theories and theoretical concepts
- demonstrate some ability to comment on secondary sources
- demonstrate some ability to select a theoretical framework that can be applied in the data analysis
- present some data analysis that draws at least partially on the chosen theoretical framework
- demonstrate some, albeit limited, evidence of independent thinking
- be relatively well expressed
- Adhere broadly to academic conventions in terms of referencing and presentation.

A mark of 40-49% (Resubmit) will be awarded to a dissertation which:

- seeks to attain some aims and objectives, although these may not be appropriate and/or feasible in all cases and may not be clearly or explicitly expressed
- displays a structure that may include some relatively important omissions or digressions
- Select a body of data that does not allow for at least partial fulfillment of the aims of the study.
- shows a limited understanding of theories and theoretical concepts
- includes a very limited discussion of secondary sources

14.1 Oral Exams

In line with University Regulation 48 relating to Higher Degrees of Masters (Taught), students must be required to present themselves for an oral examination (a 'viva voce') following submission of the MSc dissertation and as part of the assessment process. Examiners may wish to hold an oral examination for several reasons, including where this would help examiners arrive at a final mark for the dissertation, where there is uncertainty surrounding the contribution of the student to the project reported on in the dissertation (e.g., in the case of a company project), or where there are questions surrounding the authorship of the dissertation.

Students will be notified at least one week in advance of the requirement to attend an oral examination. While students will normally be required to attend the oral examination in person, it may be possible to conduct the examination remotely (e.g., via MS Teams). In all cases, all reasonable efforts will be made by the school to ensure that there is no delay to a student's graduation; however, delays may be unavoidable in some instances.

14.2 Resubmission

When the final agreed mark achieved is a grade of D (40% to 49%) then this is classed as a "fail, subject to resubmission". The extent of improvements that will be allowed for resubmission is minor corrections to the company research project or master's dissertation.

If you are required to resubmit, the markers will provide you with an agreed list of corrections to be made. You will be allowed only one further meeting with your dissertation supervisor to confirm that you understand these required changes. A form will detail the required changes. Submissions should amend this form to show how these requirements have been met by the candidate. You may be asked to highlight in colour those sections that have been modified.

The deadline for resubmission will be communicated to you following the Award Board meeting at which your dissertation (first submission) is considered. Submission arrangements for resubmission are the same as for the original submission (see section 12 above).

Your dissertation supervisor is not required to re-read the dissertation or any part thereof before resubmission. You are responsible for all activity and re-drafting to achieve the required standard.

The changes to the resubmitted dissertation are checked by a first and second marker and are capped at a mark of 55% by the Board of Examiners. This is to dissuade students seeking to gain from multiple submissions (i.e., significantly improving their mark through a second submission).

Please note that if you submit your original version in August and are permitted to resubmit, you will not be able to attend graduation in November. This is because marks must be approved by appropriate Boards of Examiners, and these do not meet in time to approve resubmitted work. Therefore, there is no gain in completing changes quickly. Ensure that your complete changes are by the stated deadlines and to the best of your ability.

Heriot-Watt University

School of Engineering and Physical Sciences

MSc Program Title:

Project Title:

Course Code:

Author:

Student full registered name :

Student HWU Matriculation Number:

Supervisor's Name:

Date:

Some useful resources

- [Home - Academic Study Skills - ISGuides at Heriot-Watt University \(hw.ac.uk\)](#)
- [technical-report-writing guide.pdf](#)
- [Advanced Word Skills \(sharepoint.com\)](#)
- [Academic Integrity and Plagiarism \(sharepoint.com\)](#)
- [Why do you need to cite and reference? \(sharepoint.com\)](#)
- [Abstracts and Executive Summaries \(sharepoint.com\)](#)
- [Synopsis? - Google Search](#)
- How to write your literature review
- <https://youtu.be/kW6Uzn-8uMI>
- [Writing a Literature Review \(sharepoint.com\)](#)
- Some helpful Video Clips:
[New Turnitin Tutorial for Canvas June 2018 – YouTube](#)
[Turnitin - Your originality report - YouTube](#)